

Dark Patterns & Consent in Healthcare Interfaces

The Psychology of Patterns, Consent in Healthcare

Cyber Psychology Portfolio Project

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Executive Summary

Consent obtained through a well-designed interface and consent obtained through a manipulative one can look identical in a compliance log: a checked box, a timestamp, a recorded “yes.” This report examines the psychological mechanisms that interface designers exploit to produce that checked box without producing genuine understanding, and uses a major, decided regulatory case to ground the analysis in real consequences rather than theory alone.

The core finding is that dark patterns do not work by deceiving people about facts; they work by engineering the moment of decision-making so that a person's fast, low-effort mode of thinking makes the choice instead of their deliberate, evaluated mode. Defaults, urgency language, and friction asymmetry (easy to accept, hard to decline) are not identical design choices; they are the mechanisms. And in a healthcare setting, where the information being consented to often involves sensitive health data, the gap between technical consent and informed consent carries both ethical and regulatory weight.

Background

In December 2022, the Federal Trade Commission reached a record settlement with Epic Games, the maker of Fortnite, requiring \$520 million in penalties, including \$275 million for children's privacy violations and \$245 million to refund consumers charged through what the FTC formally alleged were “dark patterns.”

The FTC's complaint describes several specific design choices: a counterintuitive, inconsistent button layout that leads players to incur charges from a single accidental press, a payment system that saved a link credit card by default, so a purchase just required no separate cardholder confirmation, and a refund and cancellation flow deliberately made difficult to find and complete. A related complaint further alleged that Fortnite set voice and text chat to on by default for child and team accounts without parental notice or consent, a design choice the FTC treated as a distinct unfair practice under the FTC Act.

The case matters here not because it involves healthcare but because it is one of the clearest regulatory findings that interface choices, defaults, friction placement, and button design can themselves constitute a violation, independent of whether the underlying product or service was otherwise legitimate. Internal epic communications reportedly show employees had raised concerns about these exact design choices before the FTC action, indicating the effects were foreseeable rather than accidental.

Psychology Framework

Choice Architecture and the Default Effect

Thaler and Sustein's concept of choice architecture holds that the way options are presented, particularly which option is preselected, exerts outsized influence over the outcome, independent of the options' actual merits. Status quo bias means most people accept whatever is preselected rather than actively changing it, which is precisely why a preselected "share my data" box produces far higher opt-in rates than an unchecked one, regardless of what people say they want when asked directly.

Cognitive Load and Satisficing

Herbert Simon's concept of satisficing describes how people, faced with a decision that exceeds their available attention or time, settle for an acceptable rather than optimal choice. A dance privacy notice presented at a moment of low patience produces exactly this: Users click "agree" not because they evaluated the terms, but because evaluating them is more costly than accepting the default outcome.

Dual-Process Theory and Manufactured Urgency

Kahneman's dual-process framework distinguishes fast, automatic System 1 thinking from slow, effortful System 2 evaluation. Urgency language ("Complete now," "your session expires") is particularly effective because it keeps a decision within System 1, where scrutiny is minimal, by introducing time pressure that discourages the switch to deliberate evaluation.

Loss Aversion

People weigh potential losses roughly twice as heavily as equivalent gains. Framing a consent decision around what will be lost by declining or delaying (a missed appointment, a lost future) rather than what is gained by consenting exploits this asymmetry to push acceptance, independent of the actual value of what is being agreed to.

Consent vs. Informed Consent

A sign-or-click consent is not equivalent to informed consent, which requires that the person actually understand what they are agreeing to and its consequences. The gap between the two is precisely where dark patterns operate: they are optimized to produce the former, while actively working against the conditions (time, clarity, absence of pressure). That's the latter requirement.

Fictional Application: Elite Healthcare

Applying this framework to Elite Healthcare's patient-facing portal shows how the same design choices found in the Epic Games case would function in a healthcare context, where the data at stake is health information rather than payment authorization.

Scenario

The Elite Health Care patient portal includes a data-sharing setting that allows "Care Insight," aggregated health information, to be shared with third-party wellness and disease management partners. The setting is enabled by default for every new patient account. The option exists for the client but is placed at the bottom of a long settings page and uses smaller, lower-contrast text than the surrounding context.

When a patient logs in to confirm an upcoming appointment, a banner reads, "Complete your care insight preference now to avoid delays with your visit." Most patients reading this shortly before an appointment do not want to jeopardize it. Click through the better and accept the default without locating or reading the decline option. The portal later records this interaction as informed affirmative consent with third-party data sharing.

Behavioral Analysis

The table below maps each interface element in the scenario to the dark pattern technique it represents and the psychological principle it exploits.

Interface Element	Dark Pattern Technique	Psychology Principle Exploited
Data sharing is pre-enabled at account creation	Default opt-in	Status quo bias/ default effect
Decline option in small text at the page bottom	Friction asymmetry	Satisficing under cognitive load
“Complete now to avoid delays” banner	Manufactured urgency	System 1 dominance (dual-process theory)
Delay frame as a consequence of not confirming	Lost-frame messaging	Lost aversions
Click record it as informed consent	Consent-without-comprehension	Consent vs. informed consent gap

Recommendations

Individual-Level

- Treat any pre-appointment or pre-service bundled with a sharing setting as a moment to slow down deliberately; this urgency discourages exactly that pause.
- Actively look for a decline or opt-in control rather than assuming its absence means non-existent; Dark pattern design relies on that assumption.
- Review account privacy and sharing settings periodically, outside of any time-pressured prompt, when a full system 2 evaluation is easier to engage in.

Organizational-Level

- Set data-sharing defaults to opt out rather than opt in for any use beyond direct treatment, so that the status quo bias works in favor of patient privacy rather than against it.
- Present the accept and client options with equal visual prominence in size and placement, removing the friction caused by asymmetry that the FTC specifically flagged in its Epic Games complaint.
- Remove Urgency language from consent notices, from consent and preference flows entirely, and decouple data-sharing preference from time-sensitive actions like appointment confirmation.

- Adopt layered, plain-language consent notices as a summary with the option to expand into detail rather than a single dense disclosure, and periodically audit consent flows against current FTC dark-pattern guidance.

Conclusion

The FTC's case against Epic Games established, with real financial and legal consequences, that a consent flow can be simultaneously technically valid and substantively manipulative. The same design techniques are used for default framing, friction asymmetry, manufactured urgency translated directly into a healthcare setting, where the stakes are a patient's health information rather than payment method. The Thru Line for this project and for the portfolio as a whole is that genuine consent requires evaluating not just what an interface asks but how it asks it.